AQA GCSE BIOLOGY

Infection and Response

COMPLETE RESOURCE PACK — HIGHER TIER

Chapter 3	9 Sections	Higher Tier

SECTION 1 OF 9 — REVISION NOTES WITH DIAGRAMS

SECTION 1: REVISION NOTES WITH DIAGRAMS

AQA GCSE Biology | Chapter 3: Infection and Response | Higher Tier

1.1 Communicable Diseases

A communicable disease is a disease that can be spread from one organism to another. They are caused by pathogens.

Pathogen	A microorganism that causes disease. Types include: bacteria, viruses, fungi and protists.

Communicable disease	A disease caused by a pathogen that can be spread between individuals (also called infectious disease).

Types of Pathogens:

BACTERIA	VIRUSES	FUNGI	PROTISTS
Living cells — prokaryotes	Non-living — need host cell to replicate	Eukaryotes — have cell walls made of chitin	Single-celled eukaryotes
Produce toxins that damage tissues	Replicate inside host cells, releasing new viruses and destroying host cells	Spores spread by air/water/direct contact	Many spread by vectors (e.g. mosquitoes)
Can be treated with antibiotics	Cannot be treated with antibiotics	Treated with antifungal medicines	Treated with antiprotist drugs
Examples: Salmonella, Gonorrhoea	Examples: Measles, HIV, TMV	Example: Rose black spot	Example: Malaria (Plasmodium)

How Pathogens are Spread:

- Direct contact — touching infected person or contaminated surface (e.g. gonorrhoea)
- Droplet infection — coughing, sneezing, breathing (e.g. measles, influenza)
- Contaminated food or water — ingestion of pathogens (e.g. Salmonella)
- Vectors — organisms that carry disease without being affected (e.g. mosquitoes carry malaria)
- Breaks in skin — wounds allow entry of pathogens

1.2 Viral Diseases

Measles

MEASLES — Key Facts
Pathogen: Measles virus (RNA virus)
Spread: Droplet infection — coughing and sneezing
Symptoms: Fever, red skin rash covering the body
Complications: Can be fatal — especially in malnourished children or those with weakened immune systems; can lead to pneumonia and brain damage
Treatment: No specific treatment — symptoms managed; vaccination prevents infection
Prevention: MMR vaccine (measles, mumps, rubella) — given at 12–15 months and 3–4 years

HIV and AIDS

HIV / AIDS — Key Facts
Pathogen: Human Immunodeficiency Virus (HIV) — a retrovirus
Spread: Sexual contact (exchange of body fluids); sharing contaminated needles; mother to child during pregnancy/birth/breastfeeding; blood transfusions
Mechanism: HIV attacks and destroys helper T-cells (CD4+ lymphocytes) — weakening the immune system
AIDS: Acquired Immune Deficiency Syndrome — the late stage of HIV infection when the immune system is so damaged the body cannot fight off infections or cancer
Symptoms of AIDS: Rapid weight loss, fatigue, recurring fever, night sweats, prolonged swelling of lymph nodes, opportunistic infections (e.g. tuberculosis, PCP pneumonia)
Treatment: Antiretroviral drugs (ARTs) — prevent HIV from replicating. Taken daily for life; they do NOT cure HIV but stop progression to AIDS
Prevention: Condom use; not sharing needles; screening blood donations; pre-exposure prophylaxis (PrEP)

DIAGRAM 1: HIV Infection and Progression to AIDS

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HIV INFECTION CYCLE

HIV virus                        Helper T-cell
[HIV]  ———————attaches———————▶  [T-cell]
   |                                 |
   |  injects RNA                    |
   ▼                                 ▼
Viral RNA enters T-cell     Virus uses T-cell
→ Reverse transcriptase     machinery to replicate
→ Viral DNA made            |
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  → Inserted into T-cell DNA │
     │                       ▼
     └──────────────────►  New HIV viruses bud out
                            T-cell destroyed

  Over years: T-cell count drops → immune system fails → AIDS
  ARTs block reverse transcriptase OR prevent viral assembly
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Tobacco Mosaic Virus (TMV)

TMV — Key Facts
Pathogen: Tobacco Mosaic Virus (TMV) — a plant virus
Spread: Contact between plants; contaminated tools; insects
Host: Wide range of plants including tomatoes, peppers and tobacco
Symptoms: Discoloured mosaic pattern of light and dark green patches on leaves; distorted/stunted growth; leaves may curl or turn yellow (chlorosis)
Effect: Damages chloroplasts — less photosynthesis → reduced growth and yield
Treatment/Prevention: No cure; use of virus-free seeds, crop rotation, controlling insect vectors, washing tools

1.3 Bacterial Diseases

Salmonella Food Poisoning

SALMONELLA — Key Facts
Pathogen: Salmonella bacteria (rod-shaped gram-negative bacteria)
Spread: Eating food contaminated with Salmonella bacteria (undercooked chicken, eggs, unwashed vegetables); poor food hygiene; contact with infected animals
Mechanism: Bacteria invade the gut lining AND produce toxins that cause symptoms
Symptoms: Fever, abdominal cramps, vomiting, diarrhoea (can last 4–7 days)
Treatment: Rehydration (fluids); severe cases may require antibiotics
Prevention: Thorough cooking of food (kills bacteria); good food hygiene (wash hands, surfaces); vaccination of poultry flocks in the UK

Gonorrhoea

GONORRHOEA — Key Facts
Pathogen: Neisseria gonorrhoeae (bacterium) — a sexually transmitted infection (STI)
Spread: Unprotected sexual contact; mother to baby during childbirth
Symptoms: Thick yellow/green discharge from penis or vagina; pain when urinating; many infected people show NO symptoms (asymptomatic)
Complications: If untreated — pelvic inflammatory disease (PID), infertility, increased HIV risk; in newborns — eye infections/blindness
Treatment: Antibiotics (penicillin was used; now resistant strains require different antibiotics — antibiotic-resistant gonorrhoea is a growing public health concern)
Prevention: Condom use; regular STI screening; partner notification and treatment

1.4 Fungal and Protist Diseases

Rose Black Spot (Fungal)

ROSE BLACK SPOT — Key Facts
Pathogen: Diplocarpon rosae (fungus)
Spread: Wind-dispersed spores; water splash (rain/irrigation)
Host: Roses
Symptoms: Purple/black spots develop on leaves; leaves turn yellow (chlorosis) and drop early (defoliation)
Effect: Loss of leaves reduces photosynthesis → plant weakened, reduced growth, may die
Treatment: Remove and destroy infected leaves; fungicide sprays
Prevention: Choose disease-resistant rose varieties; avoid overhead watering; improve air circulation

Malaria (Protist Disease)

MALARIA — Key Facts
Pathogen: Plasmodium (a protist — single-celled eukaryote)
Vector: Female Anopheles mosquito (the vector — carries and transmits the pathogen but is not the pathogen itself)
Spread: Mosquito bites an infected person → ingests Plasmodium → bites another person → injects Plasmodium into bloodstream
Symptoms: Recurring fever (every 48–72 hours), shaking chills, headache, muscle aches,

fatigue, sweating — cycles when Plasmodium bursts out of red blood cells
Complications: Cerebral malaria, organ failure, anaemia, death — 600,000+ deaths per year globally
Treatment: Antimalarial drugs (e.g. artemisinin-based combination therapy — ACT)
Prevention: Mosquito nets (insecticide-treated); insect repellent; antimalarial drugs (prophylaxis for travellers); draining stagnant water (mosquito breeding sites); spraying insecticides

DIAGRAM 2: Malaria Transmission Cycle

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MALARIA TRANSMISSION CYCLE

INFECTED PERSON                          MOSQUITO
┌──────────────┐                    ┌──────────────┐
│ Plasmodium   │  ← mosquito bites  │  Plasmodium  │
│ in blood     │                    │  develops in │
└──────────────┘                    │  mosquito gut│
                                    └──────┬───────┘
NEW PERSON                                 │
┌──────────────┐                           │
│ Plasmodium   │ ← mosquito bites ─────────┘
│ injected     │   and injects
│ into blood   │   Plasmodium
└──────┬───────┘
       │ Plasmodium enters liver → then red blood cells
       ▼ Red blood cells burst → fever symptoms
KEY: Mosquito = VECTOR; Plasmodium = PATHOGEN
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1.5 Human Defence Systems

The human body has three lines of defence against pathogens:

THREE LINES OF DEFENCE
LINE 1 — NON-SPECIFIC BARRIERS: Prevent pathogens entering the body
LINE 2 — NON-SPECIFIC IMMUNE RESPONSE: Attack pathogens that have entered
LINE 3 — SPECIFIC IMMUNE RESPONSE: Targeted response using lymphocytes

Line 1 — Non-Specific Physical and Chemical Barriers:

- Skin — tough physical barrier; produces antimicrobial secretions; sebaceous glands produce sebum (slightly acidic — inhibits bacterial growth)
- Mucus — lines the respiratory, digestive and reproductive tracts; traps pathogens and particles
- Cilia — tiny hair-like structures in the trachea/bronchi that wave mucus (and trapped pathogens) up and out of the airways to be swallowed
- Stomach acid — hydrochloric acid (pH 1–2) kills most pathogens swallowed in mucus or food
- Tears — contain lysozyme enzyme that destroys bacterial cell walls
- Blood clotting — platelets and clotting factors seal wounds quickly, preventing pathogen entry

Line 2 — Non-Specific Immune Response (Phagocytosis):

DIAGRAM 3: Phagocytosis
PHAGOCYTOSIS — HOW PHAGOCYTES DESTROY PATHOGENS
Step 1: Phagocyte detects chemicals released by pathogens
(chemotaxis) and moves towards the pathogen
Step 2: Phagocyte engulfs the pathogen by extending
its cell membrane around it
[Phagocyte] → [○pathogen] → engulfs
Step 3: Pathogen is enclosed in a vacuole (phagosome)
[Phagocyte: [vacuole(pathogen)]]
Step 4: Lysosomes fuse with the vacuole
→ release digestive enzymes (lysozymes)
→ pathogen is broken down and destroyed
Step 5: Phagocyte may present antigens on its surface
→ activates specific immune response (lymphocytes)
Phagocytes = neutrophils and macrophages

Line 3 — Specific Immune Response (Lymphocytes):

B LYMPHOCYTES (B-cells)	T LYMPHOCYTES (T-cells)
Produced in bone marrow; mature in bone marrow	Produced in bone marrow; mature in thymus gland
Recognise specific antigens on pathogen surface	Recognise antigens presented by infected cells/phagocytes
Produce antibodies — specific to one antigen	Helper T-cells activate B-cells and cytotoxic

	T-cells
Differentiate into plasma cells (antibody factories) and memory B-cells	Cytotoxic T-cells kill infected cells directly
Antibodies bind to antigens → agglutination, neutralisation, opsonisation	Memory T-cells remain for rapid response to reinfection

DIAGRAM 4: Specific Immune Response

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ANTIBODY-MEDIATED IMMUNE RESPONSE

Pathogen enters body
      │
      ▼
Antigens on pathogen surface detected
      │
      ▼
B-lymphocyte with complementary receptor binds to antigen
      │
      ├──► Activated by Helper T-cells
      │
      ▼
B-cell clones → Plasma cells + Memory B-cells
      │                    │
      ▼                    ▼
Antibodies produced    Remain in blood
(thousands per second) for YEARS
      │
      ▼
Antibodies bind to antigens
→ Pathogens clumped (agglutination) → destroyed by phagocytes
→ Toxins neutralised
→ Pathogens marked for destruction (opsonisation)
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Antigens and Antibodies:

Antigen	A protein or molecule on the surface of a pathogen (or cell) that triggers an immune response. Each pathogen has unique antigens.

Antibody	A protein produced by B-lymphocytes (plasma cells) that binds specifically to one type of antigen. They are Y-shaped proteins with a unique binding site (complementary to the antigen).

Memory cells	Long-lived B and T lymphocytes that remain after the primary immune response. They enable a faster, larger secondary response upon re-exposure to the same pathogen.

1.6 Vaccination

Vaccination involves introducing antigens (or weakened/dead pathogens) into the body to stimulate an immune response without causing disease.

DIAGRAM 5: How Vaccination Provides Immunity
HOW VACCINATION WORKS
Vaccine injected/ingested
Contains: dead/weakened pathogen, antigen fragments, mRNA (e.g. COVID-19 vaccines)
│
▼
Immune system responds → B-cells activated
→ Antibodies produced (primary response)
→ Memory B-cells and T-cells formed
│
▼
PERSON IS IMMUNE (no symptoms as pathogen not active)
│
Later exposure to real pathogen:
▼
Memory cells recognise antigen RAPIDLY
→ Massive secondary response within hours
→ Pathogen destroyed before symptoms develop
PRIMARY RESPONSE: slow (1-2 weeks), small antibody production
SECONDARY RESPONSE: fast (<24 hrs), large antibody production

ADVANTAGES OF VACCINATION	DISADVANTAGES / CONCERNS
Prevents disease and saves millions of lives	May cause mild side effects (soreness, fever)
Herd immunity — if enough people are vaccinated, disease cannot spread even to unvaccinated individuals	Rare severe allergic reactions (anaphylaxis) possible
Can eliminate diseases entirely (e.g.	Some people cannot be vaccinated

smallpox eradicated 1980)	(immunocompromised, allergies)
Protects vulnerable people who cannot be vaccinated	Some pathogens mutate rapidly (e.g. flu) — vaccines need updating
Cost-effective — prevention cheaper than treatment	Vaccine hesitancy — misinformation reduces uptake

HERD IMMUNITY ★
When a sufficiently large proportion of a population is immune (through vaccination or previous infection), the spread of a pathogen is greatly reduced.
Even unvaccinated individuals are protected because the pathogen cannot find enough susceptible hosts to spread.
The herd immunity threshold varies by disease: measles ~95%; polio ~80–85%; COVID-19 ~60–70%.
Herd immunity can be lost if vaccination rates fall — as seen with measles outbreaks following misinformation campaigns.

1.7 Antibiotics and Other Medicines

ANTIBIOTICS	ANTIVIRALS
Kill or inhibit the growth of bacteria	Target viruses — much harder to develop
Examples: penicillin, amoxicillin, streptomycin	Examples: antiretrovirals (HIV), oseltamivir (flu)
Work by: disrupting cell wall synthesis, protein synthesis, DNA replication in bacteria	Work by: blocking viral entry, reverse transcriptase, or assembly
Do NOT work on viruses — viruses use host cell machinery	Do not kill viruses inside host cells easily without harming host
Overuse leads to antibiotic resistance (major global threat)	Difficult because viruses mutate rapidly

Why Antibiotics Do NOT Work Against Viruses:

- Viruses are not living cells — they have no cell wall, metabolism or ribosomes of their own to target
- Viruses replicate INSIDE host cells using the host cell's own machinery
- Drugs that block viral replication would also damage host cells — making them toxic

Antibiotic Resistance — A Major Global Concern:

ANTIBIOTIC RESISTANCE
Antibiotic resistance arises when bacteria evolve (through natural selection) to survive exposure to an antibiotic.
Mechanism: Random mutations in bacteria → some individuals are resistant → antibiotics kill susceptible bacteria → resistant bacteria survive and reproduce → resistance allele increases in frequency.
MRSA (Methicillin-resistant Staphylococcus aureus) = a 'superbug' resistant to most antibiotics.
Causes of increased resistance: overprescription of antibiotics; not completing courses; agricultural overuse.
Solutions: Only prescribe when necessary; complete courses; develop new antibiotics; improve hygiene; restrict agricultural use.
IMPORTANT: Antibiotics do NOT cause mutations — they SELECT for pre-existing resistant mutants.

1.8 Development of New Medicines

Before a new drug (medicine) can be used on patients, it must go through a long process of testing to ensure it is safe and effective.

DIAGRAM 6: Drug Development and Testing Pipeline

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DRUG DEVELOPMENT AND TESTING PROCESS

STAGE 1: DISCOVERY
├── Drug identified from: plants (e.g. aspirin from willow bark;
│   digitalis from foxglove), microorganisms (e.g. penicillin
│   from Penicillium mould), or designed in a laboratory
│
STAGE 2: PRECLINICAL TESTING (in the lab)
├── Tested on cells/tissues in lab (in vitro)
├── Tested on animals (in vivo) — to check for toxicity,
│   dosage and effectiveness
│
STAGE 3: CLINICAL TRIALS (on humans)
├── Phase I: Small number of healthy volunteers
│           Test for SAFETY and side effects
│           Establish safe dose range
├── Phase II: Small group of patients with the disease
│           Test for EFFICACY (does it work?) + safety
├── Phase III: Large randomised controlled trial (RCT)
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│           Drug vs placebo (or existing treatment)
│           Double-blind: neither patients nor doctors
│           know who gets drug or placebo
│           Statistical analysis of results
│
STAGE 4: REGULATORY APPROVAL
└── Drug reviewed by medicines regulatory authority
    (e.g. MHRA in UK, FDA in USA)
    → Licensed for use if safe, effective, consistent
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The Placebo and Double-Blind Trial:

Placebo	A substance that looks identical to the drug being tested but contains no active ingredient. Used as a control to separate the drug's effect from the psychological effect of receiving treatment.

Double-blind trial	A clinical trial in which neither the patients nor the doctors/researchers know who is receiving the real drug and who is receiving the placebo — until all data is collected. Eliminates bias from both patients and researchers.

PENICILLIN — A Famous Discovery ★
Alexander Fleming discovered penicillin in 1928 when he noticed that Penicillium mould was killing bacteria in a Petri dish.
He observed a clear zone (inhibition zone) around the mould where bacteria had died — the mould was producing a chemical (penicillin) that killed bacteria.
Howard Florey and Ernst Chain developed penicillin into a usable antibiotic in the 1940s.
Penicillin works by inhibiting the synthesis of bacterial cell walls — bacteria cannot maintain their shape and burst due to osmosis.
This was one of the most important medical discoveries in history — saving an estimated 200 million lives.

1.9 Monoclonal Antibodies (Higher Tier)

HIGHER TIER ONLY ★
Monoclonal antibodies are identical antibodies produced from a single clone of B-lymphocytes.
They are all specific to the SAME antigen — so they are highly precise in their targeting.

How Monoclonal Antibodies are Produced:

DIAGRAM 7: Production of Monoclonal Antibodies
PRODUCTION OF MONOCLONAL ANTIBODIES
STEP 1: Inject mouse with specific antigen
→ Mouse produces B-lymphocytes specific to that antigen
STEP 2: Extract B-lymphocytes from mouse spleen
STEP 3: Fuse B-lymphocytes with tumour cells (myeloma cells)
→ Produces HYBRIDOMA cells
(B-lymphocyte can make antibody + tumour cell can divide indefinitely)
STEP 4: Culture hybridoma cells in laboratory
→ Divide rapidly, producing large quantities
STEP 5: Hybridoma cells produce MONOCLONAL ANTIBODIES
→ Identical antibodies, all specific to the original antigen
KEY: Hybridoma = Hybrid cell (B-cell + myeloma) that
both produces antibodies AND divides continuously

Uses of Monoclonal Antibodies:

USES OF MONOCLONAL ANTIBODIES
1. PREGNANCY TESTS: Monoclonal antibodies specific to hCG (human chorionic gonadotropin) — a hormone produced only during pregnancy. If hCG is present in urine, antibodies bind to it and trigger a colour change on the test strip.
2. MEDICAL DIAGNOSIS: Used in ELISA tests to detect specific antigens in blood/urine samples — e.g. detecting cancer markers, HIV antigens, specific pathogens.
3. CANCER TREATMENT: 'Magic bullet' therapy — monoclonal antibody attached to a cancer drug or radioactive substance. The antibody targets cancer cells specifically (binds to cancer cell antigens) delivering the treatment directly to the tumour, reducing side effects on healthy cells.
4. RESEARCH: Used in lab tests to locate or identify specific molecules in cells or tissues (immunohistochemistry).
5. TARGETED THERAPY: Herceptin (trastuzumab) — monoclonal antibody that targets HER2 receptor on some breast cancer cells, blocking signals that cause rapid cell division.

ADVANTAGES	DISADVANTAGES / CONCERNS
Highly specific — target only one type of antigen	More side effects than initially expected in clinical use
Can deliver drugs/radiation directly to cancer cells — reducing damage to healthy tissue	Expensive to produce
Used in diagnosis AND treatment	Mouse-derived antibodies can trigger immune reactions in humans — humanised antibodies being developed
Can be produced in large quantities	Takes significant time and resources to develop for each target antigen

1.10 Plant Diseases and Defences (Higher Tier)

Detecting Plant Disease:

- Visual inspection — look for discolouration, spots, wilting, distorted growth, growths/tumours on stems
- Testing with monoclonal antibodies — lateral flow tests for specific plant pathogens
- DNA testing — identifies pathogen DNA from plant samples
- Sending samples to a laboratory for microscopic analysis or culturing

Examples of Plant Diseases:

DISEASE	PATHOGEN TYPE	SYMPTOMS	SPREAD
Rose black spot	Fungus (Diplocarpon rosae)	Black/purple spots on leaves; yellowing; leaf drop	Wind and water splash
Tobacco Mosaic Virus (TMV)	Virus	Mosaic pattern (light/dark patches); stunted growth	Contact; contaminated tools; insects
Crown gall disease	Bacterium (Agrobacterium tumefaciens)	Large tumour-like growths (galls) on stems and roots	Through wounds in plant
Ash dieback	Fungus (Hymenoscyphus fraxineus)	Wilting leaves; diamond-shaped lesions on bark; death of branches	Wind-dispersed spores

Plant Defence Responses:

HOW PLANTS DEFEND THEMSELVES

PHYSICAL DEFENCES (barriers to entry):
• Cellulose cell walls — tough barrier against pathogen entry
• Waxy cuticle on leaves — waterproof layer prevents spore germination and fungal entry
• Bark — tough outer protective layer on stems; covers wounds
• Thorns and spines — deter herbivores (physical defence against animals)
CHEMICAL DEFENCES:
• Antibacterial chemicals — some plants produce chemicals that kill bacteria (e.g. mint, garlic produce antimicrobial oils)
• Poisons/toxins — some plants produce toxic compounds to deter insects and animals (e.g. nicotine in tobacco, alkaloids in deadly nightshade)
• Insect repellents — strong-smelling chemicals deter insect pests

MECHANICAL ADAPTATIONS ★
• Thorns — physical barrier against large herbivores (e.g. roses, cacti)
• Leaves that droop or curl when touched — rapid response to deter insects (e.g. Mimosa pudica)
• Hairy/sticky leaf surfaces (trichomes) — trap insects before they can damage the plant
• Deep roots — access water in drought, reducing stress that makes plants more susceptible to disease

KEY WORDS SUMMARY — Infection and Response

KEY TERM	DEFINITION
Antibiotic	A medicine that kills or inhibits the growth of bacteria (does NOT work on viruses)
Antigen	A protein/molecule on the surface of a pathogen that triggers an immune response
Antibody	A protein produced by plasma cells (B-lymphocytes) that binds specifically to one antigen
Antiretroviral	A drug that inhibits the replication of retroviruses such as HIV
B-lymphocyte	A type of white blood cell that produces antibodies; matures in bone marrow
Communicable disease	A disease caused by a pathogen that can be spread between individuals
Double-blind trial	A clinical trial where neither patient nor researcher knows who receives the drug or placebo
Herd immunity	When enough of a population is immune to prevent spread of disease even to unvaccinated individuals
HIV	Human Immunodeficiency Virus — destroys T-cells, leading to AIDS if untreated
Hybridoma	A fused cell (B-lymphocyte + myeloma tumour cell) used to produce monoclonal antibodies
Immune response	The body's response to infection — involves phagocytes and lymphocytes
Lymphocyte	A type of white blood cell involved in specific immune responses (B-cells and T-cells)
Memory cell	A long-lived B or T lymphocyte that enables rapid response to a previously encountered antigen
Monoclonal antibody	Identical antibodies produced from a single clone of B-lymphocytes; specific to one antigen
MRSA	Methicillin-resistant Staphylococcus aureus — a bacterium resistant to most antibiotics
Pathogen	A microorganism that causes disease (bacteria, virus, fungus, or protist)
Phagocyte	A white blood cell that engulfs and destroys pathogens by phagocytosis
Phagocytosis	The process by which phagocytes engulf and destroy pathogens using digestive enzymes
Placebo	An inactive substance used as a control in clinical trials to compare with the real drug

Plasma cell	A B-lymphocyte that has been activated and now produces large quantities of antibodies
T-lymphocyte	A type of white blood cell; T-helper cells activate B-cells; cytotoxic T-cells kill infected cells
Vaccination	Introducing antigens/weakened pathogens into the body to stimulate immunity without causing disease
Vector	An organism that carries and transmits a pathogen without being affected (e.g. mosquito for malaria)
Virus	A non-living particle that can only replicate inside a host cell; not affected by antibiotics

QUICK RECALL CHECKLIST

Tick each point once you can recall it confidently without notes:

☐	Name the four types of pathogens and give one example of each
☐	Describe four ways pathogens can be spread between organisms
☐	Describe the symptoms, spread and treatment of measles
☐	Explain how HIV damages the immune system and leads to AIDS
☐	Describe the symptoms of Tobacco Mosaic Virus (TMV) and explain their effect on the plant
☐	Describe the symptoms, spread and prevention of Salmonella food poisoning
☐	Explain how gonorrhoea is spread and why antibiotic-resistant strains are a concern
☐	Describe the symptoms and spread of rose black spot
☐	Explain the transmission cycle of malaria, distinguishing between pathogen and vector
☐	Describe the three lines of defence against disease in humans
☐	Explain how phagocytosis works (4 steps)
☐	Distinguish between B-lymphocytes and T-lymphocytes
☐	Explain what antigens and antibodies are and how they interact
☐	Explain how memory cells lead to a faster secondary immune response
☐	Explain how vaccination works and describe the difference between primary and secondary immune responses
☐	Evaluate the advantages and disadvantages of vaccination

☐	Explain what herd immunity is and why it requires high vaccination rates
☐	Explain why antibiotics do not work against viruses
☐	Describe how antibiotic resistance develops through natural selection
☐	Describe the stages of drug development and testing (preclinical and clinical trials)
☐	Explain the purpose of a placebo and double-blind trial
☐	Describe the discovery of penicillin by Alexander Fleming
☐	[Higher] Describe how monoclonal antibodies are produced using hybridoma cells
☐	[Higher] Give three uses of monoclonal antibodies
☐	[Higher] Describe physical and chemical defences in plants